Tensiometer



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Tensiometer

What it is used for

A tensiometer measures surface tension or interfacial tension of liquids, depending on the instrument type. Use the specified measurement method and a clean measuring probe to obtain reliable results.

Main parts

- Tensiometer main unit and display
- Force sensor or balance mechanism
- Ring, plate, or probe appropriate to the method
- Sample container and sample holder
- Calibration weights or reference standard, if specified
- Control buttons and measurement software, if available

Before you start

1. Place the instrument on a stable, level surface away from vibration and airflow.
2. Check the instrument, probe, cable, and sample holder for damage.
3. Confirm the instrument is suitable for the required surface or interfacial tension measurement.
4. Clean and dry the probe using the approved method.
5. Prepare a sufficient quantity of sample in a clean container.
6. Switch on the instrument and allow it to stabilize as recommended.
7. Verify calibration or zero setting according to the manufacturer's instructions.

How to use

1. Select the correct measurement mode and units on the instrument.
2. Install the clean ring, plate, or probe securely.
3. Place the sample container on the holder and position it correctly.
4. Set the probe height or sample stage as required by the method.
5. Zero the instrument or confirm the baseline reading.
6. Lower the probe into contact with the liquid using the prescribed procedure.
7. Start the measurement and allow the instrument to complete its reading.
8. Record the result, units, sample identification, and test conditions.
9. Repeat the measurement if required and compare the readings for consistency.
10. Clean the probe and sample holder after testing, then switch off the instrument.

⚠ Safety

- Handle glass sample containers carefully to prevent cuts and spills.
- Wear suitable gloves and eye protection when handling chemical samples.
- Keep liquids away from electrical controls and instrument connectors.
- Do not exceed the instrument's specified load or probe travel limits.
- Use only approved cleaning solvents and follow their safety instructions.

Care & storage

- Clean and dry the probe after every test using the recommended procedure.
- Store probes safely to prevent bending, contamination, or surface damage.
- Keep the instrument level and protect it from vibration and dust.
- Arrange calibration and servicing at the intervals specified by the manufacturer.

Common mistakes to avoid

- Using a contaminated or wet probe that changes the measured surface tension.
- Selecting the wrong measurement mode or units.
- Touching or moving the instrument during measurement and causing unstable readings.
- Using an unsuitable sample container or incorrect immersion depth.
- Recording readings without sample identification or test conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.